Sustainability Aware Asset Management

Portfolio Allocation with a Carbon Emission Reduction Objective

Group BR

North America + Europe | Scope 1 + Scope 2

Group Members:

Noah GRANDE
Loris ROLLANDAY
Anthony ROTTET
Guillaume VANOLLI

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Abstract

This report investigates whether carbon constraints in equity portfolio construction can be installed without sacrificing financial performance. Using 1,302 North American and European firms data from Datastream, we constructed and tested five long-only portfolios over 132 monthly observations from January 2014 to December 2024. All results are strictly out-of-sample as Portfolios are rebalanced annually each December using the preceding ten years of return data. Carbon exposure is measured using combined Scope 1 and Scope 2 emissions, expressed as a carbon footprint (tCO_2 per million USD invested) and a weighted-average carbon intensity (tCO_2 per million USD of revenue).

Two baseline portfolios are established in Part I: a value-weighted benchmark $P^{(vw)}$ and a global minimum-variance portfolio $P_{oos}^{(mv)}$. Part II introduces three carbon-constrained variants: $P_{oos}^{(mv)}(0.5)$, $P_{oos}^{(vw)}(0.5)$ and $P_{oos}^{(vw)}(NZ)$. They target respectively a 50% footprint reduction on the minimum-variance portfolio, a 50% reduction on the benchmark through a tracking-error overlay, and a 10%-per-year net-zero glide path anchored to the 2013 benchmark level.

Contents

1 Introduction

Climate considerations have gradually become a central concern in portfolio management. What was once a niche topic now sits at the heart of institutional asset allocation, driven by a combination of regulatory pressure and a growing body of academic evidence. The EU Sustainable Finance Disclosure Regulation [1] and the recommendations of the Task Force on Climate-related Financial Disclosures [2] have pushed asset managers to measure and disclose the carbon exposure of their portfolios, a requirement that barely existed a decade ago. On the academic side, Bolton and Kacperczyk [3] show that stocks with heavier carbon footprints already carry a transition risk premium: the market is pricing the possibility of future carbon taxes and regulatory tightening into current valuations. Firms exposed to abrupt shifts in consumer behavior or to stranded-asset risk face a genuine, measurable cost of capital disadvantage. However the empirical results on whether carbon constraints actually damage financial returns remain genuinely inconclusive [4]. The answer tends to depend on the investment universe, the horizon chosen, and how the constraint is implemented in practice.

This project tries to answer the problem through a rigorous, fully out-of-sample back-test of five long-only equity portfolios drawn from North American and European markets. The analysis is organized in two parts.

Part I builds two reference portfolios, a value-weighted market benchmark $P^{(vw)}$ and a global minimum-variance portfolio $P_{oos}^{(mv)}$. It then measures their risk-adjusted performance over January 2014 to December 2024. These two portfolios form the financial baseline against which the carbon aware strategies of Part II are judged.

Part II introduces three carbon-constrained portfolios based on the two previous ones, as mentioned above. The first, $P_{oos}^{(mv)}(0.5)$, asks whether a minimum-variance investor can cut their carbon footprint in half without losing too much. The second, $P_{oos}^{(vw)}(0.5)$, is aimed at the passive investor who wants to stay close to the benchmark while still reducing its footprint by 50%. The third, $P_{oos}^{(vw)}(NZ)$, follows a net-zero glide path, shrinking the footprint by 10% each year relative to the 2013 benchmark level. This strategy aligns with the Paris Agreement that a lot of large pension funds and sovereign wealth funds have adopted.

All empirical work is done in Python. Running the notebook `SAAM_Project_2026.ipynb` from top to bottom reproduces every table and figure in this report. No source data file is touched by hand: cleaning, transformation, and optimisation are all handled programmatically.

Sample convention. The data were run from end-1999 to end-2025. Following the instructions, the first portfolio allocation is set in December 2013, using the preceding ten years, i.e. January 2004 to December 2013, as the estimation window, and the last allocation is set in December 2023. Performance is then tracked month by month from January 2014 through December 2024, generating $T = 132$ monthly observations. All five portfolios are evaluated over this same 11-year window so that comparisons remain meaningful.

2 Data and Cleaning

2.1 Datasets

The raw files cover 2,545 companies worldwide. Restricting to the assigned region, `AMER` and `EUR`, gives us an initial universe of **1,302 firms** across the United States, Canada, the United Kingdom, continental Europe and Israel. For each company five different data series are used:

- The monthly total-return index $P_{i,t}$ (USD, January 2000 to December 2025), from which simple monthly returns are derived.
- The monthly market capitalisation $\text{Cap}_{i,t}$ (USD millions), used both to construct the value-weighted benchmark and to compute the carbon footprint.
- Annual Scope 1 and Scope 2 CO_2 emissions $E^{S1}_{i,Y}$ and $E^{S2}_{i,Y}$ (metric tonnes), aggregated into total emissions $E_{i,Y} = E^{S1}_{i,Y} + E^{S2}_{i,Y}$ in line with our assigned perimeter.
- Annual revenues $\text{Rev}_{i,Y}$ (thousand USD), used to compute carbon intensity.
- The one-month US risk-free rate (annualized %), converted to a monthly decimal for the calculation of the Sharpe ratio.

2.2 Cleaning Rules

Five cleaning rules are applied to the raw files as requested by the instructions. All steps are handled programmatically.

(i) Unmatched ISINs. A handful of ISINs in the files return entirely empty columns once merged on the firm identifier. These correspond to securities that Datastream could not match (likely due to corporate actions, name changes, or data gaps) and are dropped from all matrices.

(ii) Near-zero prices. Any total-return index value below 0.50 is treated as missing. Prices close to zero generate outsized returns in the following period, distorting both the covariance matrix and the back-test results.

(iii) Delisting and default. When a firm's price series ends before the close of the sample (December 2025), this signals a delisting, acquisition, or default. In such cases, a return of -100% is recorded in the first month after the last valid price. The investor is assumed to lose their entire position. This follows standard practice for avoiding survivorship bias. Dropping delisting without flagging it inflates historical returns as only firms that made it to the end are retained. Across the full sample, **112 delistings** are recorded and penalised in this way.

(iv) Simple monthly returns. Returns are computed as simple (arithmetic) monthly returns, $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$. Log returns are not used.

(v) Forward-filling of carbon and revenue data. Firms do not always publish emissions or revenue figures every year. A missing annual observation is replaced by the most recent available value (forward-fill along the time axis). This assumption is realistic and reflects an investor attitude: when lacking the data it falls back on the last known value. Any firm without *emissions* or revenue observation before year Y is excluded from the investment set for year $Y + 1$ (see Section 2.3).

2.3 Investment Set

At the end of each year $Y \in \{2013, \dots, 2023\}$ we define the set of companies eligible for the following year's portfolios. A firm enters the investment set if and only if it meets all five of the following criteria:

(i) **Region:** the Datastream `Region` field is either `AMER` (North America) or `EUR` (Europe).

(ii) **Return history:** at least 36 non-missing monthly returns over the ten-year estimation window $[Y-9, Y]$. In other words, a minimum of three years of price history. This makes sure the set has enough data to estimate the moments.

(iii) **Liquidity filter:** fewer than 50% of the monthly returns in the estimation window are exactly zero. Stale prices or illiquid prices cause the minimum variance optimizer to overweight them because of their artificially low estimated volatility retaining produce portfolios that look well behaved in sample but deteriorates out of sample.

(iv) **Valid price and market capitalisation:** a non-missing total-return index and a strictly positive market capitalisation at the December year-end date.

(v) **Carbon and revenue data:** at least one historical observation of non-missing Scope 1+2 emissions and a non-missing revenue figure for year Y itself (after forward-filling), so that the carbon footprint and carbon intensity are well-defined for every firm in the set. Importantly, **the same investment set is used across all five portfolios**: Parts I and II share an identical setup each year, meaning that differences in performance and carbon metrics can be attributed to the choice of weights alone, not to a different opportunity set.

Table 1 shows the resulting investment set sizes by allocation year.

Table 1: Investment set size by allocation year. Number of firms satisfying all five eligibility criteria at the end of year Y. The set expands over time primarily because carbon-data coverage improved markedly after 2015 as ESG reporting became widespread.

Year Y	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Firms	835	859	891	914	946	1,003	1,093	1,138	1,168	1,166	1,150

The investment set grows from 835 companies in 2013 to roughly 1,150 in the following years, driven mainly by the improvement of ESG reporting coverage [2, 5]. Before 2015, emissions reporting was largely confined to the largest company listed (via CDP and Trucost). By 2020, mid-cap coverage had improved considerably. The modest contraction in 2022–2023 is due to the firms not having fully completed the reporting for the recent years at the date of extraction.

2.4 Carbon Metrics

Two portfolio-level carbon metrics are used throughout Parts I and II.

Weighted-average carbon intensity (WACI).

$$\mathrm{WACI}_Y(\alpha) = \sum_{i \in \mathcal{S}_Y} \alpha_{i,Y} \cdot \frac{E_{i,Y}}{\mathrm{Rev}_{i,Y}/1000} \qquad (\mathrm{tCO_2 \text{ per } \$M \text{ of revenue}}). \tag{1}$$

Revenues are divided by 1,000 to convert from the reported unit (thousand USD) to millions of USD, keeping the units consistent with those of market capitalisation. WACI is the metric recommended by the TCFD [2] and the EU Paris-aligned benchmark standard [1] for expressing carbon intensity relative to economic output.

Carbon footprint (CF).

$$\mathrm{CF}_Y(\alpha) = \sum_{i \in \mathcal{S}_Y} \alpha_{i,Y} \cdot \frac{E_{i,Y}}{\mathrm{Cap}_{i,Y}} \qquad (\mathrm{tCO}_2 \text{ per \$M invested}). \tag{2}$$

The footprint measures the emissions attributable to the portfolio per million USD invested. Since the total portfolio value cancels out in calculation (ownership share $o_{i,Y} = \alpha_{i,Y} V_Y / \mathrm{Cap}_{i,Y}$, so $\sum_i o_{i,Y} E_{i,Y} / V_Y = \mathrm{CF}_Y$), the metric becomes a pure intensity measure, unaffected by portfolio size. For the value-weighted portfolio, the weights $\alpha_{i,Y} = \mathrm{Cap}_{i,Y} / \sum_j \mathrm{Cap}_{j,Y}$ reduce the footprint to the compact form $\mathrm{CF}_Y^{(vw)} = \sum_i E_{i,Y} / \sum_i \mathrm{Cap}_{i,Y}$.

CF is the metric used to define and enforce the carbon constraints in Part II, because its linearity in the portfolio weights preserves the convex quadratic structure of the optimisation problem.

3 Methodology

3.1 Estimation Window and Covariance Matrix

For each allocation year Y, the covariance matrix of returns is estimated from the $\tau = 120$ monthly returns of eligible firms over the preceding decade $[Y-9, Y]$:

$$\hat{\Sigma}_Y = \frac{1}{\tau} \sum_{k=0}^{\tau-1} (R_{t-k} - \hat{\mu}_Y)(R_{t-k} - \hat{\mu}_Y)^\top, \tag{3}$$

where $\hat{\mu}_Y$ is the sample mean vector over the same window. Expected returns play no role in any of the five portfolio optimisations. Both objectives, minimum variance and minimum tracking error rely solely on covariance estimation $\hat{\Sigma}_Y$. That's a deliberate choice, it sidesteps the estimation error problem that so often causes mean-variance optimisation to break down in practice.

When the returns are missing within the window, they are set to zero before computing (3), which guarantees that the resulting matrix is positive semi-definite. There is however a catch, with roughly 1,000 eligible firms but only 120 observations, the sample covariance matrix is rank-deficient by construction. A small ridge of $10^{-8}\mathbf{I}$ is added to recover strict positive-definiteness and allow the quadratic program solver to converge without problems.

3.2 Portfolio Optimisation Problems

All five portfolios share the same two basic constraints, they are **long-only** (weights non-negative) and **fully invested** (weights sum to one). This matters for interpretation, because no short positions are allowed, there are no negative carbon credits distorting the footprint measure. Two optimisation objectives are used across the five portfolios.

Minimum variance (MV).

$$\min_{\alpha_Y} \; \alpha_Y^\top \hat{\Sigma}_Y \alpha_Y \qquad \text{s.t.} \quad \mathbf{1}^\top \alpha_Y = 1, \quad \alpha_Y \geq 0, \quad [\text{optional: } \mathrm{CF}_Y(\alpha_Y) \leq \bar{c}_Y]. \tag{4}$$

Without a carbon constraint, this is just the standard global minimum-variance problem. Activating the cap gives $P_{oos}^{(mv)}(0.5)$ (Section 5.2).

Minimum tracking error (TE).

$$\min_{\alpha_Y} \; (\alpha_Y - \alpha_Y^{vw})^\top \hat{\Sigma}_Y (\alpha_Y - \alpha_Y^{vw}) \qquad \text{s.t.} \quad \mathbf{1}^\top \alpha_Y = 1, \quad \alpha_Y \geq 0, \quad [\text{optional: } \mathrm{CF}_Y(\alpha_Y) \leq \bar{c}_Y]. \tag{5}$$

This objective is aimed at the otherwise-passive investor who wants to stay close to the market portfolio but is willing to move the portfolio in a greener direction. Left unconstrained, the solution simply falls to the benchmark itself. Adding the carbon cap produces $P_{oos}^{(vw)}(0.5)$ and $P_{oos}^{(vw)}(NZ)$.

Because $\mathrm{CF}_Y(\alpha_Y)$ in (2) is linear in the weights, the carbon cap $\mathrm{CF}_Y(\alpha_Y) \leq \bar{c}_Y$ introduces a single linear inequality. That keeps the convex quadratic structure of both programs intact. All problems are solved numerically via the CLARABEL solver (with OSQP as a fallback) through the `cvxpy` modelling library.

3.3 Value-Weighted Benchmark

The benchmark $P^{(vw)}$ holds every eligible firm in proportion to its market capitalisation. Weights are formed using the *previous* month's capitalisations and are rebalanced monthly:

$$\alpha_{i,t}^{vw} = \frac{\mathrm{Cap}_{i,t}}{\sum_j \mathrm{Cap}_{j,t}}, \qquad R_{t+1}^{(vw)} = \sum_i \alpha_{i,t}^{vw} R_{i,t+1}. \tag{6}$$

That one detail, using prior-period weights in (6), is what keeps the construction clean: portfolio returns are constructed purely from information available at time t, with no look-ahead bias coming in.

3.4 Annual Rebalancing and Weight Drift

The four annually-rebalanced portfolios ($P_{oos}^{(mv)}$, $P_{oos}^{(mv)}(0.5)$, $P_{oos}^{(vw)}(0.5)$, $P_{oos}^{(vw)}(NZ)$) follow the same annual schedule: the optimisation runs once each December and the resulting weights are held throughout the following calendar year. Between rebalancing dates, weights drift naturally with realised returns:

$$\tilde{\alpha}_{i,t+1} = \frac{\alpha_{i,t}(1 + R_{i,t+1})}{\sum_j \alpha_{j,t}(1 + R_{j,t+1})}. \tag{7}$$

Think of a buy-and-hold investor who simply lets the market do its thing during the year. With 11 allocation dates (December 2013 through December 2023), the back-test covers exactly **132 monthly observations**, from January 2014 to December 2024.

3.5 Performance Metrics

To compare portfolios, five annualized statistics are reported during the full back-test period ($T = 132$), with r_t the monthly portfolio return and r_t^f the one-month risk-free rate:

$$\begin{aligned}
\text{Annualised average return:} \quad & \bar{R}_p^{(a)} = \bar{r} \times 12, & (8)\\
\text{Annualised volatility:} \quad & \sigma_p^{(a)} = \hat{\sigma}(r) \times \sqrt{12}, & (9)\\
\text{Annualised cumulative return:} \quad & \bar{R}_p^{(c)} = \big(1 + R_p[T]\big)^{12/T} - 1, & (10)\\
\text{Sharpe ratio:} \quad & \mathrm{SR}_p = \frac{\overline{r_t - r_t^f}}{\hat{\sigma}(r_t - r_t^f)} \times \sqrt{12}, & (11)\\
\text{Annualised tracking error:} \quad & \mathrm{TE}_p = \hat{\sigma}(r_t - r_t^{(vw)}) \times \sqrt{12}. & (12)
\end{aligned}$$

On the Sharpe ratio (11), the risk-free rate r_t^f is measured at the same monthly frequency as r_t, so that the excess return in the numerator and the volatility in the denominator are on a consistent scale before annualisation, in line with the instructions that R_f should follow the same convention as $\bar{R}_p$.

4 Part I - Standard Portfolio Allocation

4.1 Portfolio Construction

Part I constructs two portfolios: The value-weighted benchmark and the minimum-variance portfolio. The first one, $P^{(vw)}$, is rebalanced monthly using equation (6), with no optimisation involved. The second one, $P^{(mv)}_{oos}$, solves problem (4) without the carbon constraint, once each December, using only the estimated covariance matrix $\hat{\Sigma}_Y$ and the long-only, full-investment constraints. No expected-return forecast is used.

The minimum-variance approach is attractive for several reasons. First, it avoids the expected returns, that is the input most sensitive to estimation error. Second, it has consistently produced better out-of-sample Sharpe ratios than mean-variance portfolios across a wide range of universes and time periods. Finally, it's a perfect point of comparison with the value-weighted benchmark: while the market benchmark simply reflects the investors aggregate value of companies, the minimum variance strictly targets risk reduction. Since the strategy operates without any carbon consideration, its carbon footprint is purely accidental of minimizing risk.

4.2 Performance Results

Table 2 reports the annualised performance statistics for both portfolios over the full back-test period (January 2014 - December 2024).

Table 2: Part I performance statistics (January 2014 - December 2024, $T = 132$ months). Sharpe ratio uses monthly excess returns over the risk-free rate, annualised by $\sqrt{12}$.

	$P^{(vw)}$	$P^{(mv)}_{oos}$
Annualised average return	10.62%	5.96%
Annualised volatility	14.62%	14.04%
Annualised cumulative return	9.99%	5.07%
Sharpe ratio	0.72	0.42
Minimum monthly return	−13.13%	−20.20%
Maximum monthly return	12.88%	12.25%
Cumulative return (total)	184.98%	72.26%
Avg. effective # holdings	158.8	17.3

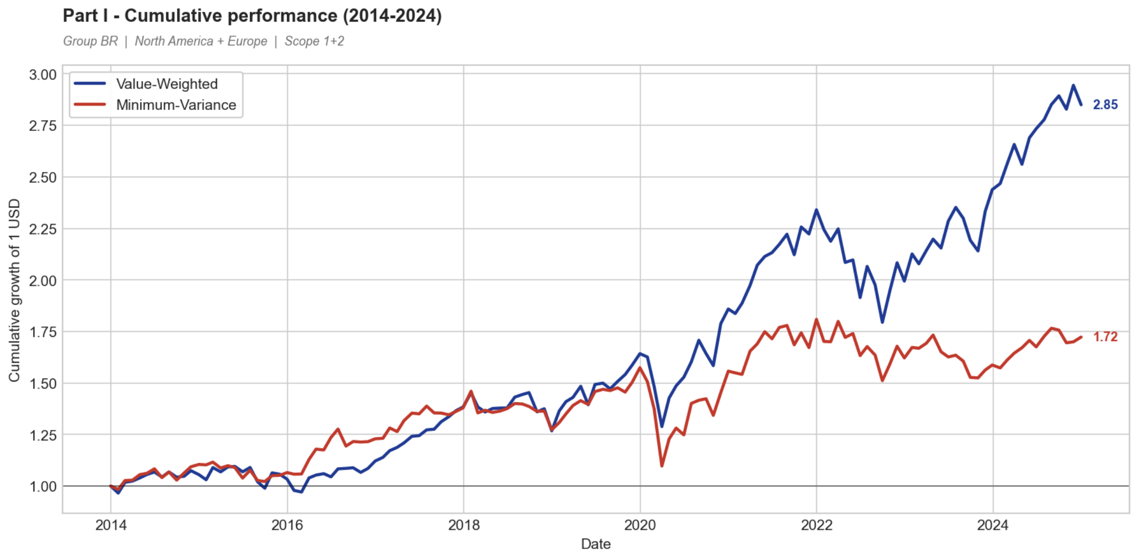

Figure 1: Cumulative performance of the Part I portfolios. Growth of one dollar invested in January 2014, plotted month by month to December 2024 (the series start at 1). The minimum-variance portfolio is more defensive in 2014-2018 but falls sharply in the March 2020 crash and never caught up. This illustrates how low-volatility investing tends to underperform during sustained market rallies.

4.3 Discussion

The benchmark dominates over this sample. The numbers are simple: $P^{(vw)}$ achieves a Sharpe ratio of 0.72 versus 0.42 for $P_{oos}^{(mv)}$, and a cumulative return of 185% against 72% over eleven years. The gap opens early and never close as Figure 1 makes it clear. This may seem surprising as the minimum-variance portfolio, by construction, minimises the variance for a given covariance estimate, and is supposed to be robust, estimation-error-resistant, a strategy built to survive. So what went wrong?

The concentration problem. The short answer is that $P_{oos}^{(mv)}$ is not a diversified portfolio. With an effective number of holdings of only $N_{\text{eff}} \approx 17$, against ≈ 159 for the value-weighted, the minimum-variance portfolio is essentially a sector bet disguised as a diversification. Since the number of assets ($\approx$1,000) is much larger than the number of return observations used for estimation (120), the sample covariance matrix is severely rank-deficient. Consequently, the optimizer capitalizes on artificial near-zero eigenvalues concentrating on firms and sectors that seemed stable during the estimation window.

In our universe, the optimizer went looking in the most stable sectors and this resulted into heavy overweight of regulated utilities such as North American and European electricity and gas companies. Although these companies offer stable, cycle-insulated cash flows and low price volatility, this investment proved costly in the 2014-2024 decade. During this decade, the companies delivering the largest return were the technology companies, the ones dominating the value-weighted benchmark, with utilities lagging behind. Moreover, the large worst-case monthly loss of $P_{oos}^{(mv)}$ (-20.2% vs -13.1% for the benchmark) explains the concentration and not diversification characteristic of the portfolio.

The technology decade. The decade 2014-2024 was exceptionally favourable for a market-cap index of North American and European equities. US technology giants, Apple, Microsoft, NVIDIA, Alphabet, Amazon and Meta, all carried massive weight in the value-weighted portfolio. As these firms experienced explosive earnings growth, they experienced surprising low volatility, driving the benchmark towards these companies. A passive investor automatically benefited from this allocation, unlike the optimizer. Relying on a 10 year look-back window, it completely misses out on this opportunity as it judges it too risky or overpriced, and therefore creates the gap.

Volatility similarity is misleading. Although both portfolios have comparable annualised volatilities (14.62% vs 14.04%), their risk profile is very different. While the benchmark's volatility represents well-diversified systematic risk, the minimum-variance portfolio is concentrated in a handful of names and therefore idiosyncratic risk. This asymmetry is clearly visible in the maximum drawdown and the worst monthly loss of $P_{oos}^{(mv)}$ during two moments in time. First, in the COVID-19 market shock in March 2020, the concentrated portfolio dropped much faster than the diversified benchmark. Diversification played a shield role and absorbed the crash. Secondly, in 2022, during the rate-hike environment, utility stock faced heavy losses as their long-duration cash-flows were discounted at higher-rates, an event that hit $P_{oos}^{(mv)}$ disproportionately.

Summary of Part I. These results establish two key benchmarks for Part II. First, $P^{(vw)}$ sets the financial standard against which carbon-constrained portfolios will be compared: 0.72 Sharpe ratio, 10.62% annualised return. Second, $P_{oos}^{(mv)}$ reveals an important and maybe counter-intuitive phenomenon that will recur in Part II: a low-volatility strategy, in this universe, is not automatically a low-carbon strategy. As we will show in Section 5.1, $P_{oos}^{(mv)}$ carries a carbon footprint nearly five times larger than the benchmark, a direct consequence of its structural overweight in utilities. This means that, for an active investor already running a minimum-variance strategy, adding carbon constraints can fix two problems simultaneously: reduce the environmental footprint *and* the idiosyncratic concentration risk.

5 Part II - Carbon-Aware Portfolio Allocation

Part II builds on Part I by adding carbon-related limits to the optimization process. We create three additional portfolios for different investor profiles and decarbonization ambitions. We first take a close look at the carbon footprint and intensity of the two baseline portfolios from Part I, which act as the reference point for the constraints that follow.

5.1 Carbon Profile of the Base Portfolios

5.1.1 Evolution of the Carbon Footprint and Intensity

For each allocation year $Y \in \{2013, \ldots, 2023\}$, the carbon footprint (CF) and weighted-average carbon intensity (WACI) of $P^{(vw)}$ and $P_{oos}^{(mv)}$ are computed using equations (2) and (1). Results are reported in Table 3 and displayed in Figure 2.

Table 3: Carbon footprint (CF) and weighted-average carbon intensity (WACI) of $P^{(vw)}$ and $P_{oos}^{(mv)}$ by allocation year (tCO_2 per million USD invested for CF; tCO_2 per million USD of revenue for WACI).

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
$P^{(vw)}$ CF	167.8	167.0	170.5	155.7	126.4	136.2	99.6	80.4	66.1	77.7	62.5
$P^{(mv)}$ CF	471.8	1291.9	1688.8	169.5	476.0	427.0	1248.7	440.0	92.7	105.5	183.2
$P^{(vw)}$ WACI	205.4	215.1	199.1	233.4	225.1	204.0	180.9	148.7	155.0	147.4	101.5
$P^{(mv)}$ WACI	936.5	893.1	782.7	174.5	519.4	663.5	995.5	549.9	320.9	220.8	109.9

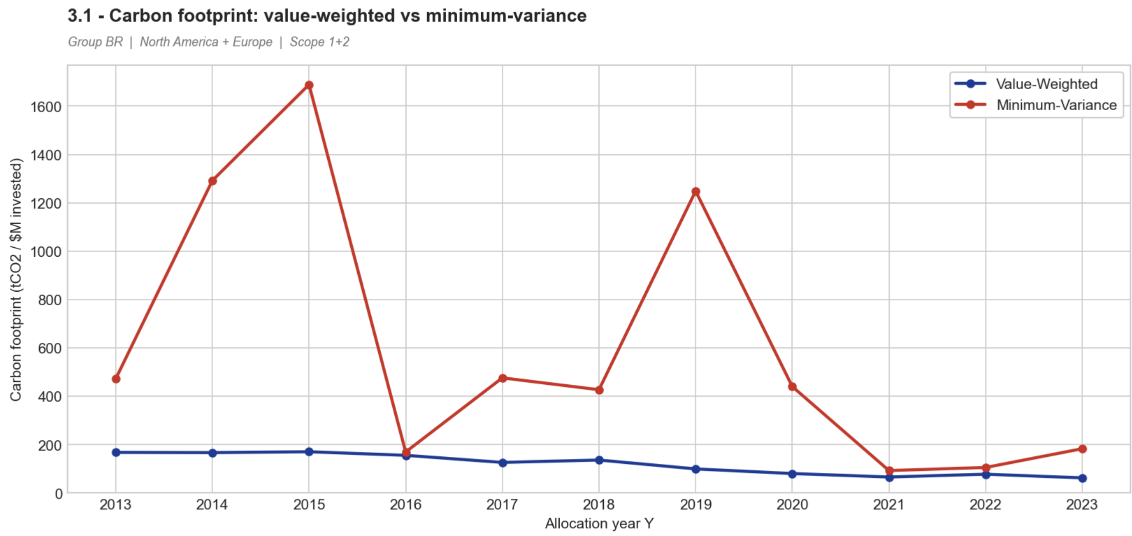

Figure 2: Carbon footprint of the base portfolios by allocation year. $P_{oos}^{(mv)}$ carries a substantially higher footprint than the benchmark in most years, with extreme spikes in 2015 and 2019. These large year-to-year swings reflect the numerical instability inherent to a concentrated portfolio built on a rank-deficient covariance matrix. The benchmark's footprint, by contrast, declines steadily over the period as the market shifts toward low-carbon technology firms.

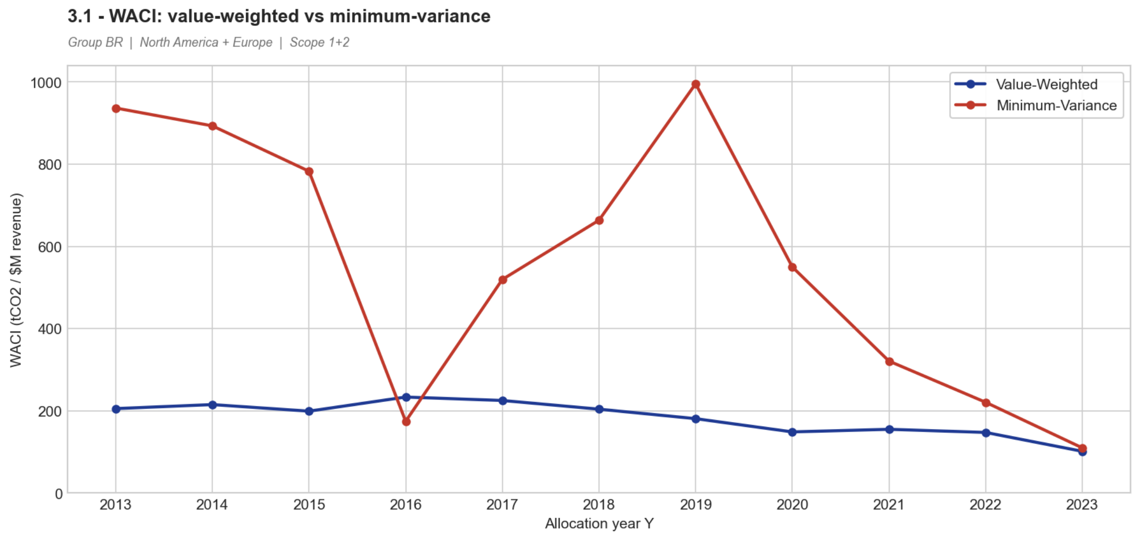

Figure 3: Weighted-average carbon intensity (WACI) of the base portfolios. Both portfolios show a declining WACI over the sample period, but $P_{oos}^{(mv)}$ starts at nearly five times the benchmark's level in 2013 and only converges in the final years as its composition becomes less extreme.

5.1.2 The Hidden Carbon Cost of Minimum Variance

The most striking finding in Table 3 is that $P_{oos}^{(mv)}$ carries on average **five times** the carbon footprint of the benchmark (599.6 vs 119.1 tCO_2/$M on average), even though both have similar levels of annualized volatility. This is a direct result of how the minimum-variance optimizer works.

As documented in Section 4, the unconstrained optimizer tilts heavily toward regulated utilities, electricity transmission, gas distribution, and water companies. These stocks have historically low and stable volatility precisely because their revenues are set by regulation, not subject to market swings. The problem is that these same sectors are at the top of the carbon intensity spectrum: thermal power generation and industrial process heat alone are responsible for a disproportionate share of total Scope 1 emissions. Chasing the quietest stocks therefore amounts to building a large implicit bet on the most carbon-heavy parts of the economy.

The important swings in $P_{oos}^{(mv)}$'s footprint (from 170 in 2016 to 1,689 in 2015, back to 170 again) are a direct symptom of the ill-conditioned covariance matrix: small changes in the estimation window lead the solver to qualitatively different optimal solutions, each implying a very different sector composition. The carbon footprint therefore inherits this instability, and its year-to-year volatility is itself a risk that the constraints introduced in Part II will help correct.

The benchmark's footprint, by contrast, falls steadily from 167.8 in 2013 to 62.5 in 2023, a reduction of over 60% in a decade. This passive decarbonisation is a byproduct of the broader market rotation toward technology, healthcare and consumer services, all sectors with low direct emissions. This trend matters for interpreting the net-zero portfolio in Section 5.4.

5.1.3 Top-10 Carbon Contributors to the Benchmark

Figure 4 identifies the ten largest contributors to the benchmark's weighted-average carbon intensity over the sample period, along with their 2023 carbon intensity and benchmark weight.

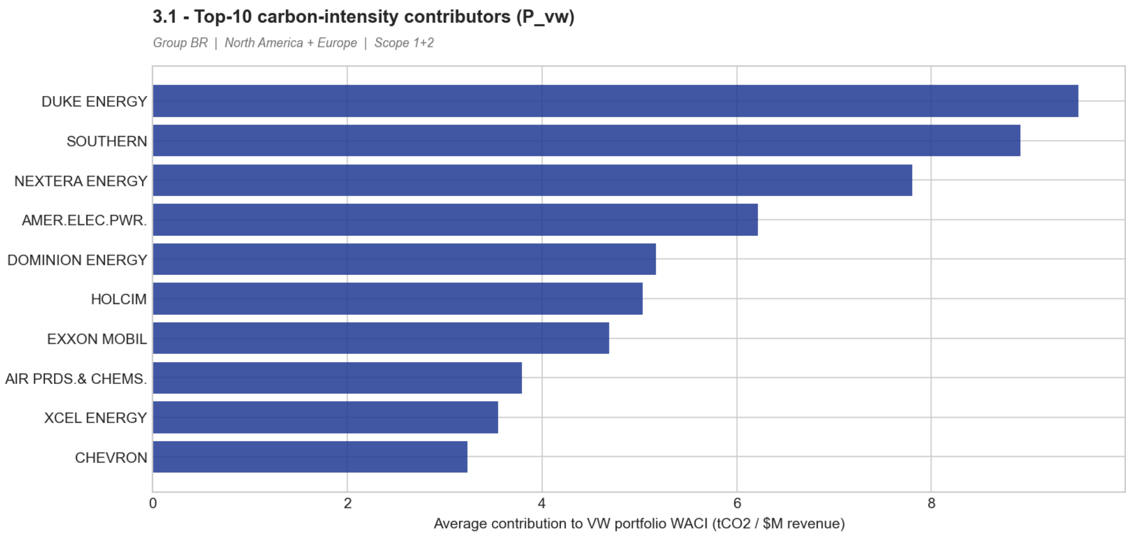

Figure 4: Top-10 average contributors to $P^{(vw)}$ WACI. Each bar represents the firm's average annual contribution to $P^{(vw)}$'s WACI (tCO_2/\$M of revenue), computed as the product of its benchmark weight and its carbon intensity, averaged over 2013–2023. A small number of utilities and energy companies account for a disproportionate share of the benchmark's carbon footprint, and are the primary targets of any decarbonisation strategy.

This concentration has a concrete implication: decarbonizing the portfolio does not require a broad restructuring. Trimming a small number of high-intensity names is enough to remove a large part of the footprint with limited impact on diversification and tracking error. This is the core idea behind the efficiency of passive strategies in Sections 5.3 and 5.4.

5.2 Minimum Variance with a 50% Footprint Cut

5.2.1 Construction

$P_{oos}^{(mv)}(0.5)$ solves the minimum-variance problem (4) with one additional constraint: in each year Y, the portfolio's carbon footprint must not exceed 50% of $P_{oos}^{(mv)}$'s footprint in that same year:

$$\mathrm{CF}_Y(\alpha_Y) \leq 0.5 \times \mathrm{CF}_Y(P_{oos}^{(mv)}). \tag{13}$$

The target is set relative to $P_{oos}^{(mv)}$'s *out-of-sample* footprint rather than a fixed absolute level, so that the 50% reduction remains meaningful regardless of how the unconstrained portfolio's footprint evolves year to year.

Before solving, we check feasibility: the long-only, fully-invested constraint imposes a minimum achievable footprint of $\bar{c}_Y^{\min} = \min_i E_{i,Y}/\text{Cap}_{i,Y}$ (i.e. the footprint of the least carbon-intensive firm in the investment set). If 50% of $P_{oos}^{(mv)}$'s footprint falls below this floor, the problem has no solution and we relax the cap to $\bar{c}_Y^{\min} \times 1.0001$. In practice, **all 11 annual targets proved strictly feasible**: the 50% level remained always above the long-only minimum, so no relaxation was triggered.

5.2.2 Results

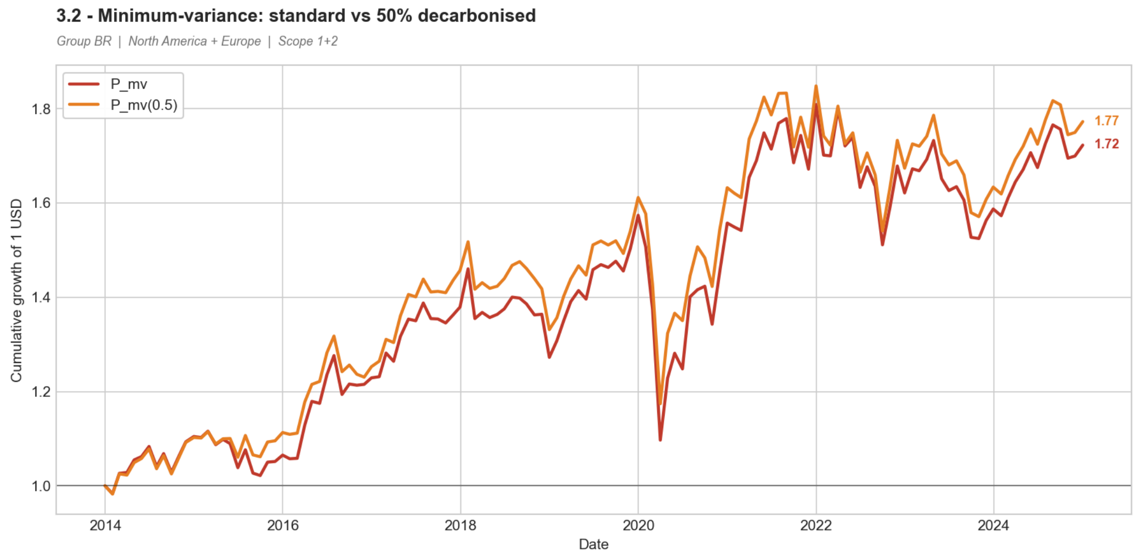

Figure 5: Cumulative performance: $P_{oos}^{(mv)}$ vs $P_{oos}^{(mv)}(0.5)$. The carbon constraint modestly improves cumulative performance over the sample. The gap widens after 2019, as the constrained portfolio avoids some of the heaviest utility positions that weigh on the unconstrained version during periods of sector rotation.

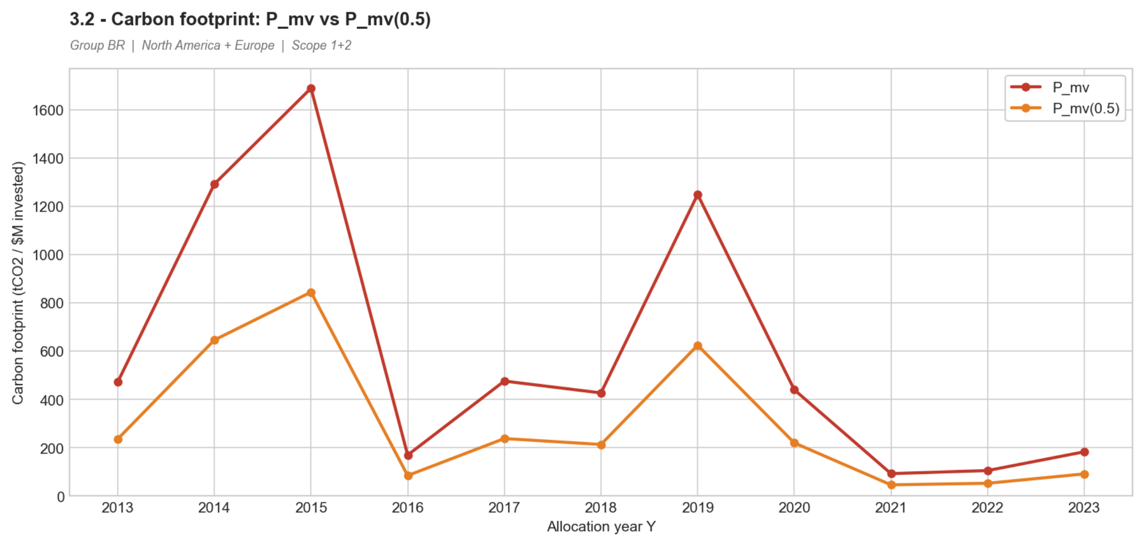

Figure 6: Carbon footprint: $P_{oos}^{(mv)}$ vs $P_{oos}^{(mv)}(0.5)$. The constrained portfolio's footprint is exactly half the unconstrained portfolio's in each year, confirming that the carbon constraint binds throughout. The large spikes of 2015 and 2019 are fully absorbed and the year-to-year volatility of the footprint is eliminated.

Table 4: Performance comparison: $P_{oos}^{(mv)}$ vs $P_{oos}^{(mv)}(0.5)$ (January 2014 – December 2024).

	$P_{oos}^{(mv)}$	$P_{oos}^{(mv)}(0.5)$
Annualised average return	5.96%	6.07%
Annualised volatility	14.04%	13.03%
Annualised cumulative return	5.07%	5.34%
Sharpe ratio	0.42	0.46
Minimum monthly return	−20.20%	−17.76%
Maximum monthly return	12.25%	12.71%
Cumulative return (total)	72.26%	77.25%
Annualised tracking error	9.66%	8.84%
Avg CF (tCO_2/\$M)	599.6	299.8
Avg WACI (tCO_2/\$M rev)	560.6	255.6
Avg effective # holdings	17.3	17.6

5.2.3 Discussion

Adding a carbon constraint produces a Pareto improvement: the carbon footprint falls by 50% while every financial metric improves. The Sharpe ratio rises from 0.42 to 0.46, annualised volatility drops from 14.04% to 13.03%, the worst monthly drawdown shrinks from −20.2% to −17.8% and cumulative return increases from 72.3% to 77.3%. The cost of decarbonization is here strictly negative.

This result is counter-intuitive but economically coherent. The unconstrained optimiser tilts toward carbon-heavy utilities because they appear stable over the estimation window, but as argued in Section 4, this stability is a statistical artefact of rank-deficiency rather than a genuine out-of-sample property. Forcing the optimiser away from the most carbon-intensive names therefore also pushes it away from its most concentrated and fragile positions. The effective number of holdings barely changes (17.3 vs 17.6), but the composition improves: the portfolio moves away from the most extreme utility overweights toward a slightly broader set of low-volatility names with lower carbon intensity.

The takeaway is simple: for a minimum-variance investor, carbon risk and concentration risk are two sides of the same coin. The stocks that inflate the footprint are the same ones that drive up the portfolio's specific risk. Cutting one reduces the other.

5.3 Tracking-Error Minimisation with a 50% Cut

5.3.1 Construction

$P_{oos}^{(vw)}(0.5)$ targets the "otherwise-passive" investor: an asset manager who would naturally hold the market portfolio but faces a mandate to reduce its carbon footprint. The objective is to stay as close as possible to $P^{(vw)}$ by minimising tracking error (equation (5)), subject to halving the benchmark's carbon footprint each year:

$$\mathrm{CF}_Y(\alpha_Y) \leq 0.5 \times \mathrm{CF}_Y(P^{(vw)}). \tag{14}$$

As with $P_{oos}^{(mv)}(0.5)$, all 11 annual targets proved strictly feasible.

5.3.2 Results

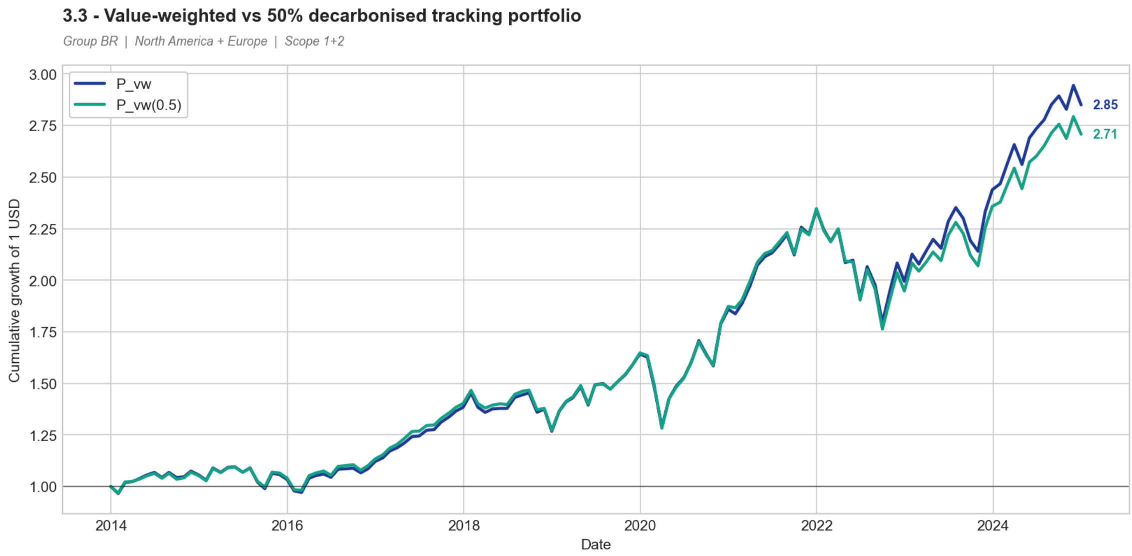

Figure 7: Cumulative performance: $P^{(vw)}$ vs $P^{(vw)}_{oos}(0.5)$. The two return paths are nearly indistinguishable throughout the sample. With an annualised tracking error of 1.09%, the constrained portfolio shadows the benchmark closely in every month, a negligible deviation by any institutional standard.

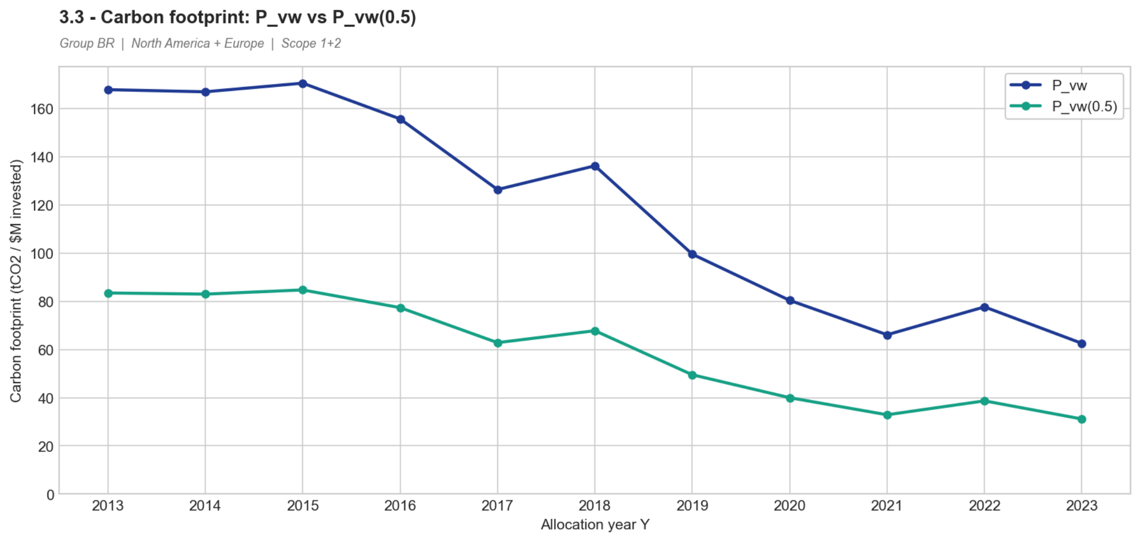

Figure 8: Carbon footprint: $P^{(vw)}$ vs $P^{(vw)}_{oos}(0.5)$. The constrained portfolio's footprint is permanently halved relative to the benchmark. As the benchmark decarbonises naturally over the decade, both series decline in parallel, maintaining the 50% gap throughout.

Table 5: Performance comparison: $P^{(vw)}$ vs $P^{(vw)}_{oos}(0.5)$ (January 2014 – December 2024).

	$P^{(vw)}$	$P^{(vw)}_{oos}(0.5)$
Annualised average return	10.62%	10.16%
Annualised volatility	14.62%	14.63%
Annualised cumulative return	9.99%	9.48%
Sharpe ratio	0.72	0.69
Minimum monthly return	−13.13%	−14.18%
Maximum monthly return	12.88%	12.91%
Cumulative return (total)	184.98%	170.74%
Annualised tracking error	—	1.09%
Avg CF (tCO_2/\$M)	119.1	59.2
Avg WACI (tCO_2/\$M rev)	183.2	90.7
Avg effective # holdings	158.8	157.8

5.3.3 Discussion

Reducing the benchmark's carbon footprint costs only **0.03 of Sharpe ratio** (0.69 vs 0.72) and **1.09% of annualised tracking error**. The portfolio remains broadly diversified (157.8 effective holdings vs 158.8 for the benchmark) and closely replicates the benchmark's risk-return profile. For a passive investor, decarbonisation is nearly free in this universe and over this period.

This connects directly to the top-10 analysis of Section 5.1: the benchmark's footprint is driven by a small number of high-intensity names. Reducing their weight modestly and redistributing across the remaining ≈980 eligible firms is enough to halve the footprint. Since these high-intensity names carry relatively small benchmark weights, the tracking error induced is minimal. The portfolio ends up slightly underweighting utilities and energy, and modestly overweighting the UK, Sweden and Israel, geographic shifts small enough to have no material bearing on performance.

This also reveals a key asymmetry between the two investor types. For $P^{(mv)}_{oos}(0.5)$, a 50% cut off an already very high base still leaves an average CF of 299.8, well above the benchmark's 119.1. For $P^{(vw)}_{oos}(0.5)$, the same relative cut produces the lowest average CF of all five portfolios (59.2). The starting point matters: an overweight of carbon-heavy utilities is much harder to decarbonise efficiently than a diversified market exposure where carbon is concentrated in a few easily-trimmed names.

5.4 Net-Zero Portfolio

5.4.1 Construction

$P^{(vw)}_{oos}(NZ)$ uses the same tracking-error objective as $P^{(vw)}_{oos}(0.5)$ but replaces the fixed 50% cap with a progressively tightening carbon glide path. The ceiling is anchored to the 2013 benchmark footprint ($\mathrm{CF}^{(vw)}_{2013} = 167.8$ tCO_2/\$M) and tightens by $\theta = 10\%$ per year:

$$\mathrm{CF}_Y(\alpha_Y) \leq (1-\theta)^{Y-2013+1} \times \mathrm{CF}^{(vw)}_{2013}, \qquad \theta = 10\%. \tag{15}$$

The resulting ceilings run from 151.0 in 2013 down to 52.7 in 2023, a cumulative reduction of roughly 69% from the 2013 baseline. This structure follows the Paris-aligned frameworks adopted by major institutional investors, notably the Net Zero Asset Managers initiative, which prescribes a minimum annual decarbonisation rate of 7–10% to stay consistent with a 1.5°C temperature pathway.

All 11 annual targets proved strictly feasible, as the minimum long-only footprint was below the glide-path ceiling in every year.

5.4.2 Results

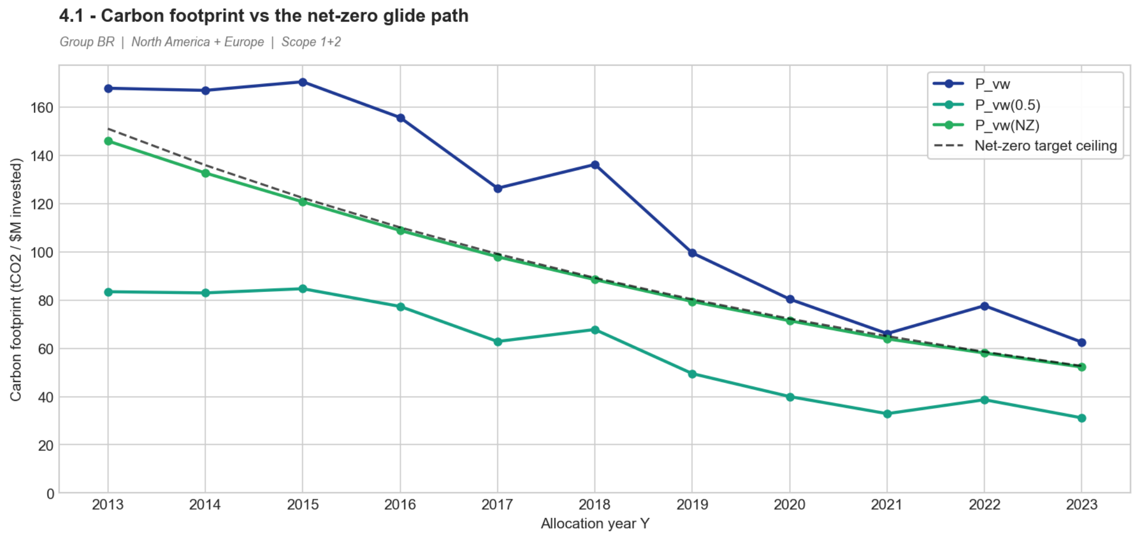

Figure 9: Carbon footprint and the net-zero glide path. $P_{oos}^{(vw)}(NZ)$ closely tracks the glide-path ceiling (dashed) in every year. The fixed-50% portfolio $P_{oos}^{(vw)}(0.5)$ falls *below* the glide path in later years, because "50% of the contemporaneous benchmark" becomes a more demanding target than the 2013-anchored ceiling once the benchmark has naturally decarbonised.

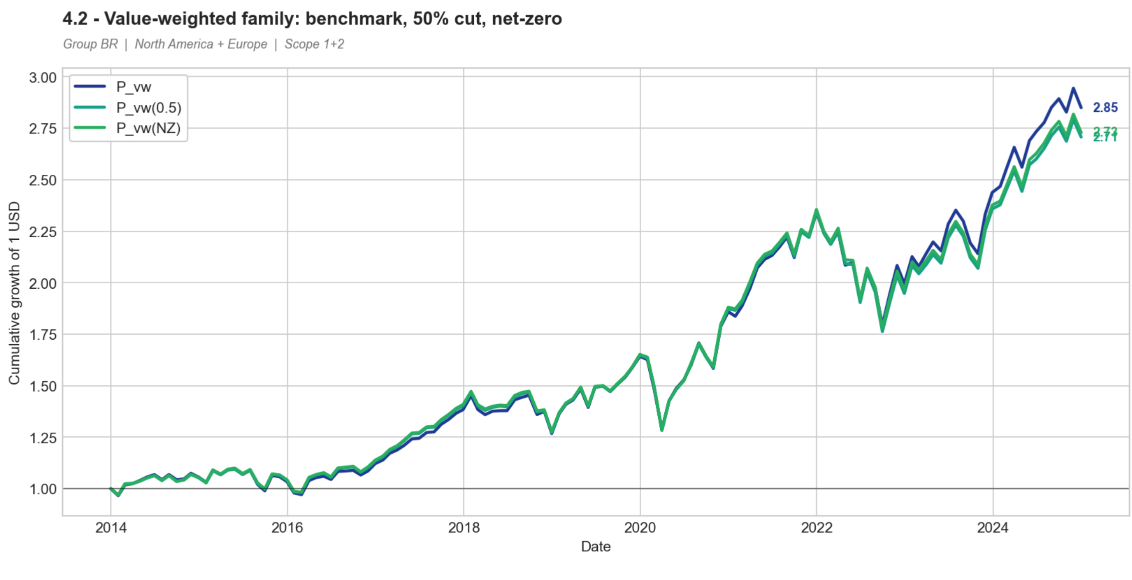

Figure 10: Cumulative performance of the value-weighted family. The three portfolios $P^{(vw)}$, $P_{oos}^{(vw)}(0.5)$ and $P_{oos}^{(vw)}(NZ)$ track each other closely throughout the sample. Despite their very different carbon constraint designs, all three converge to the same performance region, confirming that the financial cost of decarbonisation is small for a passive investor in this universe.

Table 6: Performance comparison: $P^{(vw)}$, $P^{(vw)}_{oos}(0.5)$ and $P^{(vw)}_{oos}(NZ)$ (January 2014 – December 2024).

	$P^{(vw)}$	$P^{(vw)}_{oos}(0.5)$	$P^{(vw)}_{oos}(NZ)$
Annualised average return	10.62%	10.16%	10.23%
Annualised volatility	14.62%	14.63%	14.64%
Annualised cumulative return	9.99%	9.48%	9.56%
Sharpe ratio	0.72	0.69	0.69
Minimum monthly return	−13.13%	−14.18%	−14.19%
Maximum monthly return	12.88%	12.91%	12.90%
Cumulative return (total)	184.98%	170.74%	172.90%
Annualised tracking error	—	1.09%	1.13%
Avg CF (tCO_2/\$M)	119.1	59.2	92.7
Avg WACI (tCO_2/\$M rev)	183.2	90.7	130.3
CF vs $P^{(vw)}$	—	−50%	−22%
Avg effective # holdings	158.8	157.8	161.4

5.4.3 Discussion

$P^{(vw)}_{oos}(NZ)$ achieves a 22% average reduction in carbon footprint relative to the benchmark (92.7 vs 119.1 tCO_2/\$M), with a Sharpe ratio of 0.69 and tracking error of 1.13%, nearly identical to $P^{(vw)}_{oos}(0.5)$. Its cumulative return of 172.9% marginally exceeds $P^{(vw)}_{oos}(0.5)$'s 170.7%, as the glide-path constraint is less binding in early years and keeps the portfolio closer to the benchmark.

One result may seem paradoxical at first: $P^{(vw)}_{oos}(NZ)$'s average footprint (92.7) is actually higher than $P^{(vw)}_{oos}(0.5)$'s (59.2), despite being a net-zero strategy. The explanation lies in the anchoring: the glide path is fixed to the 2013 benchmark level, but the benchmark itself decarbonised by over 60% over the decade. By 2023, the cumulative −69% ceiling relative to 2013 translates into only about −16% relative to the contemporaneous benchmark, a far less demanding target than the fixed −50% rule. A more rigorous net-zero design should either anchor the glide path to a rolling benchmark or combine the cumulative rule with a floor such as "at most 50% of the current benchmark footprint". This point is revisited in Section 7.

6 Consolidated Results

Table 7 brings together the performance and carbon metrics of all five portfolios.

Table 7: Consolidated comparison of all five portfolios (January 2014 – December 2024, $T = 132$ months). Annualised performance statistics, ex-post tracking error against the benchmark, sample-average carbon footprint and WACI, footprint change relative to $P^{(vw)}$, and average effective number of holdings $N_{\text{eff}} = 1/\sum_i \alpha_i^2$. Carbon figures in tCO_2 per million USD (invested for CF, of revenue for WACI).

	$P^{(vw)}$	$P_{oos}^{(mv)}$	$P_{oos}^{(mv)}(0.5)$	$P_{oos}^{(vw)}(0.5)$	$P_{oos}^{(vw)}(NZ)$
Ann. average return	10.62%	5.96%	6.07%	10.16%	10.23%
Ann. volatility	14.62%	14.04%	13.03%	14.63%	14.64%
Ann. cumulative return	9.99%	5.07%	5.34%	9.48%	9.56%
Sharpe ratio	0.72	0.42	0.46	0.69	0.69
Min monthly return	−13.1%	−20.2%	−17.8%	−14.2%	−14.2%
Max monthly return	12.9%	12.3%	12.7%	12.9%	12.9%
Cumulative return	184.98%	72.26%	77.25%	170.74%	172.90%
Ann. tracking error	—	9.66%	8.84%	1.09%	1.13%
Avg CF (tCO_2/\$M)	119.1	599.6	299.8	59.2	92.7
Avg WACI (tCO_2/\$M rev)	183.2	560.6	255.6	90.7	130.3
CF vs $P^{(vw)}$	—	+404%	+152%	−50%	−22%
Avg effective N	158.8	17.3	17.6	157.8	161.4

6.1 The Cost of Decarbonisation

The results reveal a clear and consistent pattern across all five portfolios. For the **benchmark-aware investor**, the cost of decarbonisation is negligible: both $P_{oos}^{(vw)}(0.5)$ and $P_{oos}^{(vw)}(NZ)$ achieve meaningful carbon reductions (−50% and −22% respectively) at a cost of only 0.03 Sharpe ratio points and approximately 1.1% tracking error. The portfolios remain highly diversified (157–161 effective holdings) and virtually indistinguishable from the benchmark in cumulative performance. For the **active minimum-variance investor**, the cost of decarbonisation is *negative*: $P_{oos}^{(mv)}(0.5)$ improves on every financial metric while halving the carbon footprint.

Three structural factors explain why decarbonisation is so cheap in this setting:

Carbon exposure is highly concentrated. A handful of utilities and energy companies account for the bulk of both portfolios' footprints (Figure 4). Trimming a small number of high-intensity names suffices to achieve a large aggregate reduction, while leaving the vast majority of the portfolio unchanged.

Decarbonising and de-risking are aligned for the min-variance investor. The carbon-heavy utilities that $P_{oos}^{(mv)}$ over-weights are exactly the concentrated, numerically-unstable positions that drive its poor out-of-sample performance. The carbon cap corrects a genuine portfolio construction flaw.

The benchmark decarbonised on its own. The value-weighted footprint fell by more than 60% between 2013 and 2023 as the index rotated toward technology and services. Any glide path anchored to 2013 levels is therefore progressively easier to satisfy, which explains why both $P_{oos}^{(vw)}(0.5)$ and $P_{oos}^{(vw)}(NZ)$ always found feasible solutions.

Caveat: these conclusions are sample-specific. The 2014–2024 decade was an extended bull market in which low-carbon technology leaders happened to be the best-performing assets. In a regime where energy, materials and utilities outperform, as they did briefly in 2022, a carbon constraint would create a real and measurable performance drag. The figures here should be read as the cost of decarbonisation in a favourable climate-transition regime, not as a universal free lunch.

7 Limitations

1. **Scope 3 emissions are excluded.** A first limitation of our study is that our analysis covers only Scope 1 and Scope 2 emissions, as assigned to our group. For many sectors, financial companies, oil majors, retailers, Scope 3 (value-chain emissions) represent the largest share of their carbon footprint, often exceeding 70%. Excluding these emissions likely understates the true carbon exposure of both benchmark and the constrained portfolios and potentially misses the transition risk installed in the supply chain.

2. **Estimation risk in the covariance matrix.** With approximately 1,000 eligible firms and only 120 monthly observations, the sample covariance matrix is severely rank-deficient. The minimum-variance portfolio exploits artificial near-zero eigenvalues and concentrates in ≈17 effective names, with a carbon footprint that swings between 170 and 1,689 tCO_2/\$M from one year to another. Applying covariance shrinkage methods, such as Ledoit-Wolf [6] or factor models, would reduce estimation error and stabilise the minimum-variance portfolio. Implementing these techniques represents the single most impactful methodological improvement available.

3. **Transaction costs and turnover are not modeled.** Currently, transaction costs and portfolio turnover are not modeled, all back-tests assume frictionless annual rebalancing. In practice, given that carbon intensities fluctuate, carbon-constrained portfolios may exhibit higher annual turnover than the benchmark. One company can be clean one year and become dirty the next year. Incorporating realistic execution costs could partially erode the financial advantage of $P^{(vw)}_{oos}(0.5)$ and $P^{(vw)}_{oos}(NZ)$.

4. **Net-zero glide-path design.** The glide path in $P^{(vw)}_{oos}(NZ)$ is anchored to the 2013 benchmark footprint and tightens by a fixed $\theta = 10\%$ per year. However, the benchmark decarbonised by 60% over the 10-year window. The ceiling becomes progressively less restrictive regarding the actual market and is in 2023 only 16% cleaner than the market. To create a more ambitious portfolio, we could anchor to a rolling benchmark or add a 50% of current benchmark floor but we risk that the constraint becomes mathematically infeasible without introducing short selling, carbon offsets, or a cash allocation.

5. **Carbon metric design and data quality.** The carbon footprint uses market capitalisation in the denominator. A panic market movement mechanically inflates the carbon intensity even when the real emissions physically remain the same. Transitioning to Enterprise Value Including Cash (EVIC), the EU regulatory standard, would remove this artificial volatility and bias. Additionally, the carbon data remains uneven. Coverage is poor before 2015, the most recent year reporting is often incomplete, Scope 1+2 are self reported with different methodologies. Consequently, the use of forward-filling to report missing data propagates stale observations, potentially adding more noise to the carbon metrics.

8 Conclusion

On a North American and European equity universe, back-tested monthly over the 10-year period 2014-2024, we constructed a value-weighted benchmark, a global minimum-variance portfolio, and three carbon constrained variants. Results came up with the minimum-variance portfolio underperforming the benchmark by 4.7 percentage points per year and carrying a carbon footprint five times larger. This occurred because low-volatility investing in this universe implicitly overweights carbon-heavy regulated utilities. However, imposing a carbon constraint corrected the imbalance with minimal financial cost. For the passive investor, a tracking-error overlay achieved a 50% carbon reduction at a marginal cost of 0.03 in the Sharpe ratio and a 1.1% tracking-error while a 10%-per-year net-zero glide path cost barely more. For the active minimum-variance investor, decarbonisation actually improved performance because it caused the model to break up its dangerous high-carbon positions. Across all models, the carbon constraint was always feasible without sacrificing diversification.

The efficiency of decarbonisation in this setting rests on three conditions specific to our sample period: carbon exposure is concentrated in a few specific stocks, these companies have small market-cap weights in the benchmark, and the sample period happened to reward technology assets over energy stocks, lower in carbon footprints. Future research should test whether these conclusions hold true in tougher periods, notably in a period of high energy prices and inflation, and whether more ambitious decarbonisation targets, such as rolling benchmarks or Scope 3 integration, remain financially viable.

Use of Large Language Models (LLMs)

During this project, we used Claude (Anthropic) and Gemini (Google) as coding and writing assistants. Specifically, these tools were used to: debug Python error messages and solver configurations; suggest cleaner approaches to data manipulation and polish the grammar and readability of the final report. We explicitly confirm that all core methodological choices (investment set criteria, optimisation formulations, carbon metric definitions, back-testing logic), all numerical results, and all interpretation of findings are entirely our own work. Our group remains fully responsible for the correctness and academic integrity of this submission.

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